D 11801 **(Pages : 4)** **Name.....**

Reg. No.....

THIRD SEMESTER (CUCBCSS–UG) DEGREE EXAMINATION
NOVEMBER 2021

B.Sc. L R P (Alternate Pattern)

A11—BASIC NUMERICAL SKILLS

(2014—2016 Admissions)

Time : Three Hours Maximum : 80 Marks

Part A

Answer **all** *questions.*
Each carries 1 *mark.*

I. Choose the correct answer :

1 $x = 4 + 8y$ is ————————.

(a) Quadratic. (b) Linear.

(c) Exponential. (d) None of the above.

2 When A = {1, 2, 3} its power set has ———————— Elements.

(a) 2. (b) 4.

(c) 6. (d) 8.

3 When all observations occur with equal frequency ———————— is ill defined.

(a) Mean. (b) Median.

(c) Mode. (d) None.

4 Histogram is a ————————.

(a) Graph. (b) Diagram.

(c) Pictogram. (d) None.

5 ———————— is a one-dimensional diagram.

(a) Bar Diagram. (b) Line Diagrams.

(c) Both a & b. (d) None of the above.

Turn over

Fill in the blanks :

6 Value of the determinant of $\begin{vmatrix} 2 & 4 \\ 8 & 2 \end{vmatrix}$ is ————————.

7 What sum of money invested at 4% Simple interest will amount to Rs. 510 in 6 months ————————.

8 Find Harmonic Mean. 3, 4, 5, 6 ————————.

9 In the relation R = $\{(1,1)(4,2)(16,4)\}$ Range of the relation is ————————.

10 If A = $\{1, 2, 3\}$ and B = $\{1, 2, 3\}$ then A $\cup$ B is ————————.

(10 × 1 = 10 marks)

Part B

Answer any **eight** *questions.*
Each carries 2 *marks.*

11 If A = $\{1, 4, 7, 10\}$ B = $\{2, 4, 5, 8\}$ U = $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ Find A' $\cap$ B.

12 Represent A' by using venn diagram

13 If A = $\{5, 8\}$ B = $\{4, 5\}$ Find B × A.

14 Find A + 4B :

$$A = \begin{pmatrix} 2 & 3 & 1 \\ 0 & -1 & 5 \end{pmatrix} B = \begin{pmatrix} 1 & 2 & -1 \\ 0 & -1 & 3 \end{pmatrix}.$$

15 Find the value of determinants :

$$A = \begin{vmatrix} 5 & 7 & 2 \\ 2 & 3 & 1 \\ 4 & 6 & 2 \end{vmatrix}.$$

16 Solve the equation $y^2 - 10y + 9 = 0$.

17 Solve $8x + 7y = 10$
$11x + 10y = 10$.

18 n^{th} term of a sequence is given by $a_n = 4_n + 7$. List the first *four* terms of the sequence.

19 Find the compound interest for Rs. 7000 for 4 years if interest is payable half yearly at 6% p.a.

20 Calculate geometric mean from the following figures 57.5, 87.75, 53.5, 73.5, 81.75.

(8 × 2 = 16 marks)

Part C

Answer any **six** *questions.*
Each carries 4 *marks.*

21 What do you mean by Lorenz Curve ? What are its uses ?

22 Define Index numbers. Explain the characteristics of Index numbers.

23 Calculate the mean and standard deviation for the following table giving the age distribution of 542 members :

Age in years	...	20-30	30-40	40-50	50-60	60-70	70-80	80-90
No. of members	...	3	61	132	153	140	51	2

24 State the differences of Mean Median and Mode. How do you choose an appropriate average.

25 Prove that $A \cap (A \cup B) = A \cup (A \cap B)$ by means of Venn diagram.

26 Solve $9x + 3y - 4z = 35$
$x + y - z = 4$
$2x - 5y - 4z = -48$.

27 Find the 10th term of GP $-\frac{3}{4}, \frac{1}{2}, -\frac{1}{3}, \frac{2}{9}$.....

28 Given that $A + B = \begin{pmatrix} 2 & 5 \\ 7 & 8 \end{pmatrix}$ $A - B = \begin{pmatrix} 6 & 8 \\ 4 & 3 \end{pmatrix}$. Find 2A.

(6 × 4 = 24 marks)

Turn over

Part D

Answer any **two** *questions.*
Each carries 15 *marks.*

29 Out of 1200 students appeared for B.Com examination, 750 failed in accountancy, 600 failed in Business organization and 600 failed in costing. 450 failed in both accountancy and business organization, 400 failed in both accountancy and costing, 150 failed in business organizations and costing and 20 failed in all subjects. How many students passes all three subjects ?

30 Solve the following equations by using matrices :

$5x - 6y + 4z = 15$
$7x + 4y - 3z = 19$
$2x + y + 6z = 46.$

31 Find Fisher's Index Number from the following data :

Commodity	2005		2008	
	Price	*Quantity*	*Price*	*Quantity*
A	2	74	3	82
B	5	125	4	140
C	7	40	6	33

(2 × 15 = 30 marks)